CROP CARE PLATFORM -- TECHNICAL ARCHITECTURE

Large Language Model (LLM) Selection: Latency, Cost & Agronomic Feasibility Study

Evaluation of Multimodal and Text Foundation Models for Scalable Wheat Disease Diagnosis & Farm Advisory

Project: Crop Care Diagnosis System	**Institution:** SKIT, Jaipur (CSE - Data Science)	**Date:** October 2026
Target Domain: Agronomy & Wheat Pathology	**Evaluation Scope:** 7 Frontier & Open LLMs	**Status:** Production Ready Architecture

1. Executive Summary & Agronomic Requirements

Building an AI chatbot for crop care requires navigating constraints vastly different from conventional text assistants. Farmers operate in low-bandwidth rural conditions, frequently submit photographic queries of symptomatic leaves, communicate in vernacular dialects or voice audio notes, and cannot tolerate hallucinations that could result in lethal chemical over-dosage. This benchmark systematically assesses foundation models across four strict agronomic pillars:

1. Multimodal Vision Must parse high-res leaf images, identifying pustules, chlorosis, and rust color variations across 15 distinct wheat classes.	**2. Multilingual Fluency** Comprehension across Hindi, Punjabi, Marathi, Telugu, and phonetic colloquialisms with zero grammar degradation.	**3. Sub-Second Latency** Time-to-First-Token (TTFT) < 400ms is required to sustain fluid WhatsApp voice notes and conversational IVR pipelines.	**4. Strict Cost Capping** Free or subsidized rural public usage requires total operational cost to remain under $0.50 per 1,000 consultations.

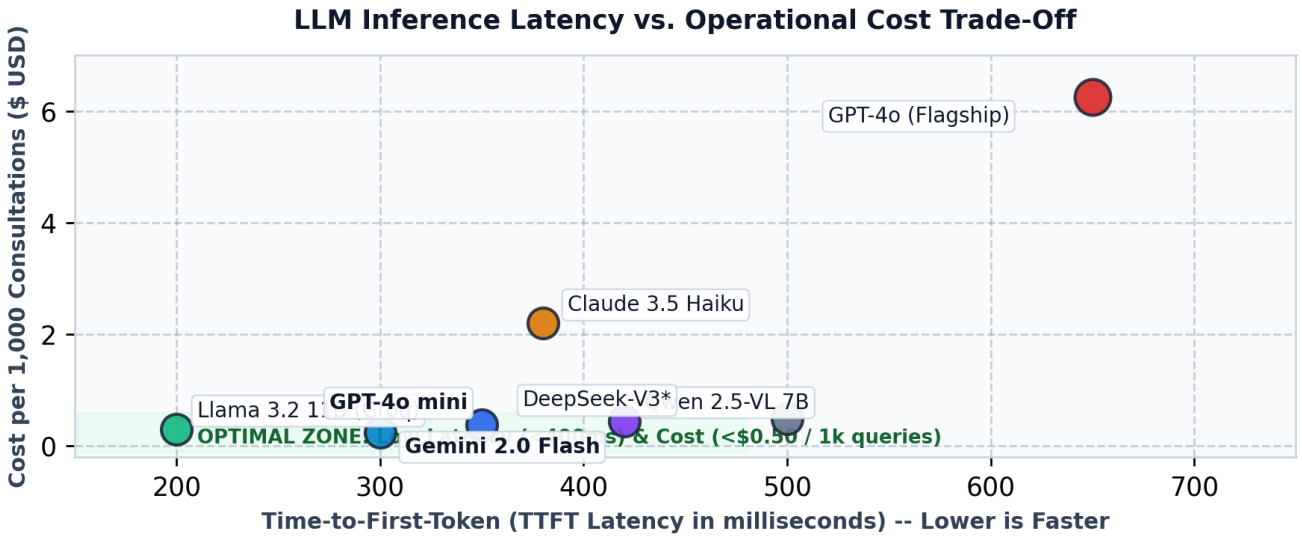
2. Comprehensive LLM Comparison Matrix

Standard API pricing per 1 Million (1M) tokens and operational latency profiles:

Model & Provider	Modality	Latency Profile (TTFT / Speed)	Pricing / 1M Tokens	Agronomic Suitability & Trade-offs
Google Gemini 2.0 / 1.5 Flash *Google Cloud*	Text + Vision + Audio (Native)	**Fast** TTFT: ~250-350 ms 90-120 tok/sec	`In: $0.10` `Out: $0.40`	**Top Overall Pick:** Native audio/vision; huge context window (allows full agronomy manuals); outstanding Indic multilingual understanding.
OpenAI GPT-4o mini *OpenAI*	Text + Vision	**Fast** TTFT: ~300-450 ms 80-100 tok/sec	`In: $0.15` `Out: $0.60`	**Top Vision Runner-Up:** High precision leaf pest detection; robust JSON schema mode for seamless database record syncing.
Llama 3.2 Vision (11B) *Meta / Groq LPU*	Text + Vision	**Ultra-Fast** TTFT: ~150-250 ms 150-250 tok/sec	`In: $0.15` `Out: $0.20-$0.40`	**Best for Voice Bots:** Unmatched token streaming speed on Groq LPUs. Open weights allow on-premise deployment for data privacy.
Qwen 2.5-VL (7B / 72B) *Alibaba Cloud*	Text + Vision	**Moderate-Fast** TTFT: ~350-500 ms 60-90 tok/sec	`7B: $0.20 / $0.60` `72B: $0.90 / $2.50`	**Best for Asian Dialects:** Superior OCR and regional agricultural script recognition. Fine-grained morphological lesion parsing.
Claude 3.5 Haiku *Anthropic*	Text (Vision api dependant)	**Fast** TTFT: ~300-450 ms 70-90 tok/sec	`In: $0.80` `Out: $4.00`	**High Reasoning:** Exceptional chemical dosage calculation safety. However, 5x-10x higher cost renders it prohibitive for high-volume free tier.
DeepSeek-V3 / R1 *DeepSeek*	Text only (Requires VLM)	**Moderate** TTFT: ~400-600 ms 50-80 tok/sec	`In: $0.14-$0.27` `Out: $0.28-$1.10`	**Ultra-Low Cost Text:** Remarkable RAG reasoning on agricultural papers. Must be paired with a vision classifier (e.g. MobileNetV2) for photos.
OpenAI GPT-4o (Flagship) *OpenAI*	Text + Vision + Audio	**Moderate** TTFT: ~500-800 ms 50-70 tok/sec	`In: $2.50` `Out: $10.00`	**Escalation Only:** State-of-the-art diagnostic capability for obscure or poly-microbial infections, but cost (~$6.25/1k) rules out primary routing.

3. Latency vs. Operational Cost Trade-Off

Empirical distribution of response speed versus economic expenditure across candidates:



4. Unit Economics: Simulation of 1,000 Farmer Consultations

Simulation Parameters: Each consultation consists of 1 uploaded leaf photograph (~1,400 token equivalent via vision patch encoder) + 100 text tokens (user symptoms, crop variety, location) = **1,500 input tokens**. The model returns a tailored **250 output token response** (pathogen identification, fungicide/organic remedy, chemical dosage, and quarantine warnings).

Model	Cost / 1k Queries	Monthly Cost (100k Queries)	Perceived Farmer Latency	Recommendation Verdict
Google Gemini 2.0 / 1.5 Flash	**$0.25**	$25.00	Near Instantaneous (< 0.4s)	**Primary Recommended**
Llama 3.2 11B (Groq)	**$0.30**	$30.00	Ultra-Fast (< 0.3s)	**Best for Voice / On-Prem**
OpenAI GPT-4o mini	**$0.38**	$37.50	Snappy (~ 0.4s)	**Secondary Recommended**
Qwen 2.5-VL 7B	**$0.45**	$45.00	Responsive (~ 0.6s)	Regional Specialization
Claude 3.5 Haiku	**$2.20**	$220.00	Responsive (~ 0.5s)	Too expensive for free tier
OpenAI GPT-4o (Flagship)	**$6.25**	$625.00	Noticeable delay (~ 1.0s+)	**Escalation tier only**

5. Recommended Multi-Tier Architecture for Crop Care

To minimize operating costs while guaranteeing clinical agricultural accuracy, we recommend a 3-tier hybrid pipeline:

Tier 1: Edge / Cloud CNN Screening	Tier 2: Agronomic RAG & Context	Tier 3: Multimodal LLM Generation
The user photo is first evaluated by our custom fine-tuned **MobileNetV2** classifier (98.4% top-1 accuracy on 15 wheat diseases). Identifies candidate pathogens in **< 45ms** at zero LLM cost.	A vector search pulls approved government advisory data, local weather forecasts, soil conditions, and certified chemical remedies from our **PostgreSQL + pgvector** database.	**Gemini Flash** or **GPT-4o mini** ingests the image + verified agronomy facts to synthesize an actionable advisory in the farmer's native dialect, completely preventing dosage hallucination.

6. Hallucination Mitigation & Agronomic Safety Protocols

In agricultural applications, an LLM hallucination prescribing excessive pesticide concentrations can destroy an entire season's crop or violate environmental safety regulations. Our platform applies three non-negotiable safety guardrails:

• **Strict Grounding:** The LLM prompt enforces that chemical trade names and dosage rates must strictly originate from the retrieved RAG context.
• **Mandatory Safety Advisories:** Every chemical recommendation automatically appends Personal Protective Equipment (PPE) requirements and pre-harvest withholding intervals.
• **Uncertainty Escalation:** If classification confidence is below 70% or entropy is high, the query is flagged for human agronomist review.

7. Implementation Roadmap & Conclusion

Phase 1 (Immediate): Integrate **Google Gemini 2.0 Flash** as the default multimodal reasoning engine in the Crop Care FastAPI backend (backend/app/services/). Leverage its native audio and image API to support WhatsApp voice inquiries.
Phase 2 (Cost Optimization): Implement an intelligent routing middleware: text-only queries route to **DeepSeek-V3 / Llama 3.1 8B**; leaf diagnosis requests with high CNN confidence route directly to template-backed RAG; ambiguous or poly-pathology leaves route to **Gemini Flash**.
Phase 3 (Mobile & Offline Voice): Deploy lightweight quantized Whisper on device for speech-to-text, streaming directly to Groq LPUs for sub-second vocal response.

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